

Raghav Senthil Kumar

623-332-2168 | raghavs9@illinois.edu | linkedin.com/in/raghav | github.com/raghav

EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Statistics

Expected: May 2029

- **Coursework:** Computer Science I (Java), Computer Science II (C++), Data Science & ML (Python), Calculus III
- **Activities:** Disruption Lab, SIGRobotics, AI Alignment, Agentic AI, BuildIllinois, Innovation LLC

EXPERIENCE

Founding Software Engineer | Clusion

September 2025 – January 2026

- Participated in the design and implementation of a *Python (FastAPI)* document search engine for debate research on *AWS* as a member of an *Agile* team of **10 engineers**, serving **10K+ site visitors**, from design to deployment
- Reduced index bloat by using a de-duplication algorithm to flag the **30%** of evidence documents that were identical
- Engineered a text-retrieval algorithm that validates the source URLs and removes statistical outliers with **IQR**
- Optimized retrieval across **600K+ documents** in *Elasticsearch* to return search results in **under 1 second**

Product Development Intern | Jetson

January 2025 – February 2025

- Contributed to the execution of key tasks via **7 iterations of UI/UX testing** on **3 experimental features**
- Provided input on product strategy in a **5-page report** to optimize user engagement, delivered to Jetson's CEO
- Reworked user onboarding in *Figma* to go from **12 to 5 pages**, verified through **A/B testing** on 30 test users

Data Analytics Intern | STEM E

June 2024 – July 2024

- Supported content strategy by analyzing **3 years** of TikTok data (**200+ posts**) in *Python (pandas)* and *Excel*
- Determined that “science fun facts” content generated **69.3%** of views despite being only **7%** of total output
- Proposed a “fun facts” strategy to the Executive Director, who adopted it and saw engagement climb **24%**

PROJECTS

EcoBin | [Research Paper](#) | [Source Code](#)

March 2026 – May 2026

- Prototyped an embedded smart waste bin on *Arduino Uno* that sorts waste with under **2 seconds** of latency
- Developed a waste classifier with a two-stage *EfficientNetV2-S* architecture (**4.39M** parameters), trained in *PyTorch* on **15K+ images** to reach **96.13% accuracy** across 4 disposal pathways, then published a paper on *arXiv*
- Automated a synthetic data pipeline using *U2-Net* segmentation and alpha-compositing that produced **9K** images
- Won **2nd at Arizona State Science Fair** and published **2 original datasets** on *Kaggle* from the project

SchoolPulse AI | [Live Website](#) | [Source Code](#)

June 2026

- Developed **3 edge AI tools** that capture a school's energy, water and food consumption data for later analysis
- Wrote the *FastAPI* backend (*Python*) that feeds water, waste and energy data from **3 tools** into **4 Random Forest models** for anomaly detection, surfacing actionable steps via a *Gemma* Voice LLM on a *Next.js* dashboard
- Managed a **5-member team** from initial ideation to presentation as a **USAII Global Hackathon Finalist**

Chronos | [Live Website](#) | [Source Code](#)

December 2025

- Engineered a **5-step Python algorithm** to scan calendar availability and return a ranked list of time slots with email confirmation per booking on Google Calendar, validated across **14 unit and integration test cases**
- Developed a *React/Firebase* agent that books meetings for large groups (30+ ppl) straight into Google Calendar
- Coded a host-controlled meeting lobby that authenticates invited participants and queues follow-up meetings

Truck Pricing VLM | [Live Website](#) | [Source Code](#)

September 2026

- Led a **4-member team** on a pricing algorithm for **Kamion (YC S22)**; **2nd at UIUC Founder's Hackathon**
- Specified a *JSON schema* that makes a *Claude Sonnet 4.6* agent return condition, brand, model and year per photo
- Reworked the schema through failed predictions until **10 consecutive images** were correct across **3 features**
- Developed a *quantile regression* price model from extracted features and validated **87% accuracy** on **30 samples**

SKILLS

Programming Languages: C++, Python, Java, JavaScript, SQL, HTML/CSS

Frameworks/Platforms: FastAPI, Flask, Node.js, React, AWS, Firebase, PostgreSQL, Google Cloud

DevTools: Arduino IDE, Clang, Catch2, Git, GitHub, Docker, CI/CD, VS Code, Claude Code

AI/ML: OpenCV, PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face, Claude API, RAG, GenAI, LLMs

Concepts/Practices: Embedded Software, Software Design, Object Oriented Programming, Full Stack Development